Solution Set 1

Discrete Structures

7th day of September of the year of our Lord 2026

1. For the sake of simplicity and brevity, we will refer to whatever universe of discourse we are talking about at the moment as Ω for every part of this problem.

   a. The sentence $\forall x(\neg(x = x))$ is *true* precisely when the Ω contains no objects. In order to see this, we will argue both directions of this biconditional separately.

   In the *"forward"* direction, we will argue that $\forall x(\neg(x = x))$ implies Ω is empty. First, let's assume $\forall x(\neg(x = x))$.[1] In order to argue that Ω is empty, let's assume *towards a contradiction* that Ω is *not* empty.[2] Since Ω is not empty, it contains an object; let's call this object t. Recall, from our axioms about equality, that $\forall x(x = x)$. Since this is a universal statement, it should apply to t in particular; we can therefore say $t = t$. For the same reason, since we believe the universal claim $\forall x(\neg(x = x))$, we can say $\neg(t = t)$ in particular. However, now we've said both $t = t$ and $\neg(t = t)$; that's clearly a contradiction. ↯
   Since the assumption that Ω was not empty lead us to a contradiction, it should be reasonable to conclude that Ω really is empty.

   In the *"backward"* direction, we will attempt to argue that if Ω is empty, then $\forall x(\neg(x = x))$. Let's begin by assuming Ω is empty. In a similar manner to our argument above, assume *towards a contradiction* that $\forall x(\neg(x = x))$ is *false.*[3] That means *not all* objects x make the statement $\neg(x = x)$ *true…* so there must be *some* object that makes $\neg(x = x)$ *false!* Let's call that object t. We now know that t exists in Ω, and that t has the property $\neg(t = t)$. However… the existence of t as an object in Ω contradicts our initial assumption that Ω was empty! ↯
   Since the assumption that $\forall x(\neg(x = x))$ was *false* led us to a contradiction, we can conclude that $\forall x(\neg(x = x))$ is actually *true.*

   Therefore, $\forall x(\neg(x = x))$ is *true* if and only if Ω is empty.

   b. We claim that $\exists x \forall y(x = y)$ is *true* if and only if Ω contains exactly one object.

   In the *"forward"* direction, let's assume $\exists x \forall y(x = y)$ is *true.* The fact that this sentence claims *"There exists an x with the property…"* means Ω must contain at least one object—let's call it t_1.[4] To argue that Ω does not contain more than one object, we assume towards a contradiction that Ω contains at least one *other* object—let's call it t_2. Since this is *another* object, distinct from t_1, that should imply $\neg(t_1 = t_2)$, right? Now, recall that t_1 has the property that $\forall y(t_1 = y)$. Since that is a universal statement, it should apply to t_2 in particular, so that $t_1 = t_2$. Now we have a contradiction: we can't have both $t_1 = t_2$ and $\neg(t_1 = t_2)$ at the same time! ↯
   Our initial assumption that Ω contained more than one object was clearly mistaken. We therefore concluce Ω contains *exactly* one object.

   In the *"backward"* direction, let's assume Ω contains exactly one object, and let's give it the name t. We can now verify that t has the property $\forall y(t = y)$ by going through each object y in Ω and verifying that $t = y$ is *true.* The first object we

[1] Making this assumption should make sense because we are trying to show that $\forall x(\neg(x = x)) \rightarrow$ "Ω is empty."

[2] We have added an additional assumption to our argument, which we should be extremely wary of. In fact, we have assumed ***the opposite of our intended conclusion.*** While we usually are not allowed to make additional unwarranted assumptions in a logical argument, we are making this specific assumption—the opposite of what we are trying—to try to arrive at a *contradiction!* That's why we made the *"towards a contradiction"* comment earlier. The idea is this: if we can show that the assumption that *"Ω is not empty"* leads to a contradiction, then that assumption must be *false,* and hence its *negation* (which is the statement *"Ω is empty"* that we originally wanted to justify) must be *true.* This is the classic framework of a *"proof by contradiction,"* which is formalized by *Negation Introduction.*

[3] Notice that, although we are going to try a *"proof by contradiction"* style argument just as before, we could take a different approach. For example, it would be reasonable to argue that $\forall x(\neg(x = x))$ must be *true* in the empty universe because we interpret the $\forall$ symbol as iterating through each object x in our universe of discourse and asserting $\neg(x = x)$ about each one of them; if each possible value of x makes the statement $\neg(x = x)$ *true,* then $\forall x(\neg(x = x))$ must be *true.* We now go through all of the objects in Ω … oh look, we're done. We have now checked each object in Ω. (How many objects did we check? None!) Since every single one of the objects we checked made the statement $\neg(x = x)$ *true,* we conclude $\forall x(\neg(x = x))$ is *true.*

[4] In fact, we can now argue that Ω contains *exactly* one object very simply by observing that the property that t_1 has is $\forall y(t_1 = y)$, which says precisely that *everything in the universe is equal to t_1.* The main text presents a slightly more detailed alternative to this perfectly adequate argument.

encounter in Ω is t; we now observe $t = t$ is *true* because we have an axiom that says $\forall x(x = x)$. Oh, there are no more objects in Ω to check (because Ω contains *exactly* one object). Since $t = y$ was *true* for every object y in Ω, as we just verified, we are justified in saying $\forall y(t = y)$. Since t is an example of an object in Ω with the property $\forall y(t = y)$, we can conclude $\exists x \forall y(x = y)$ is *true.*

Therefore, $\exists x \forall y(x = y)$ is *true* if and only if Ω contains exactly one object.

c. We claim that the sentence $\exists x \exists y(\neg(x = y) \wedge \forall z(x = z \vee y = z))$ is *true* if and only if Ω contains exactly two objects. We argue why below in two parts.

In the *"forward"* direction, assume $\exists x \exists y(\neg(x = y) \wedge \forall z(x = z \vee y = z))$ is *true.* That means there exist two objects—let's call them t_1 and t_2—with two distinct properties: first, $\neg(t_1 = t_2)$, and second, $\forall z(t_1 = z \vee t_2 = z)$. The first property, which says $\neg(t_1 = t_2)$, immediately tells us that t_1 and t_2 are *distinct* from each other (*i.e.,* they are *different objects*). That obviously implies Ω contains at least two objects. Now, the second property says $\forall z(t_1 = z \vee t_2 = z)$; in English, this means everything in Ω is equal to either t_1 or t_2. So, hypothetically, if we had some mysterious third object t_3 in Ω, we would know that either $t_1 = t_3$ is *true* or that $t_2 = t_3$ is *true.* Regardless of which one is *true,* it means that t_3 is one of the two objects we are already familiar with, and therefore we can't have a *distinct* third object in Ω. Therefore, Ω contains exactly two objects.

In the *"backwards"* direction, let's assume Ω contains exactly two objects, and let's call them t_1 and t_2. That immediately means $\neg(t_1 = t_2)$ is *true;* otherwise, there would actually be *less* than two objects. Additionally, because there are *no more* than two objects, any other object in Ω must actually be either t_1 or t_2. Formally, we can express this by saying $\forall z(t_1 = z \vee t_2 = z)$. Since we have asserted two properties about t_1 and t_2 at the same time, we should be justified in saying $\neg(t_1 \wedge t_2) \wedge \forall z(t_1 = z \vee t_2 = z)$. Further, since this sentence is *true* about the two particular objects t_1 and t_2 in Ω, we can conclude that $\exists x \exists y(\neg(x = y) \wedge \forall z(x = z \vee y = z))$ is *true.*

Therefore, we conclude that $\exists x \exists y(\neg(x = y) \wedge \forall z(x = z \vee y = z))$ is *true* if and only if Ω contains exactly two objects.

2. Consider the binary logical connectives $\overline{\wedge}$ and $\overline{\vee}$ defined by the truth table below.

p	q	$p \mathbin{\overline{\wedge}} q$	$p \mathbin{\overline{\vee}} q$
$\top$	$\top$	$\bot$	$\bot$
$\top$	$\bot$	$\top$	$\bot$
$\bot$	$\top$	$\top$	$\bot$
$\bot$	$\bot$	$\top$	$\top$

Table 1. A truth table defining the binary connectives $p \mathbin{\overline{\wedge}} q$ and $p \mathbin{\overline{\vee}} q$ that are commonly known as *"NAND"* and *"NOR"* respectively.

Either one of these logical connectives would be enough to affirm the claim that every proposition can be equivalently expressed using only one single binary logical connective. We will justify this by demonstrating—with an appeal to truth tables—how any expression involving any of the five logical connectives $\neg$, $\wedge$, $\vee$, $\rightarrow$, and $\leftrightarrow$ can be rewritten using only the $\overline{\vee}$ connective we just introduced.[5]

[5] Notice that, based on some of the observations given in this problem, we actually only really need to take care of translating $\neg$ and $\vee$; however, we will be going through all of them for the sake of demonstration.

First, we take care of the $\neg$ unary connective on its own. We will demonstrate with a truth table that $\neg p \equiv p \mathbin{\overline{\vee}} p$. In order to see this more clearly, we should first realize that $p \mathbin{\overline{\vee}} q \equiv \neg(p \vee q)$; we will add individual columns for each of the expressions $p \vee p$ and $\neg(p \vee p)$ to make this point clear. Remember: the way to interpret a truth table is that the columns on the left, which we show here without colors shaded in, represent the *input* truth values to an expression for each propositional variable, and the columns on the right that are shaded in represent the *outputs* for each corresponding expression based on those inputs.

p	$\neg p$	$p \vee p$	$\neg(p \vee p)$	$p \mathbin{\overline{\vee}} p$
$\top$	$\bot$	$\top$	$\bot$	$\bot$
$\bot$	$\top$	$\bot$	$\top$	$\top$

Table 2. A truth table showing that $\neg p \equiv p \mathbin{\overline{\vee}} p$ for any proposition p. In order to see this, simply observe that the column labelled by $\neg p$ is identical to the column labelled by $p \mathbin{\overline{\vee}} p$.

This shows us that, in order to negate a proposition p, we can write $p \mathbin{\overline{\vee}} p$. Now, if the idea that $\overline{\vee}$ is the negation of $\vee$ makes sense, it should be clear how we can translate a disjunctive expression like $p \vee q$: construct $p \mathbin{\overline{\vee}} q$, which represents $\neg(p \vee q)$, and then *negate* that proposition by applying the translation we just discovered in the previous table. This will show us that $p \vee q \equiv (p \mathbin{\overline{\vee}} q) \mathbin{\overline{\vee}} (p \mathbin{\overline{\vee}} q)$. Again, for clarity, we will show an intermediate step computing $p \mathbin{\overline{\vee}} q$.

p	q	$p \vee q$	$p \mathbin{\overline{\vee}} q$	$(p \mathbin{\overline{\vee}} q) \mathbin{\overline{\vee}} (p \mathbin{\overline{\vee}} q)$
$\top$	$\top$	$\top$	$\bot$	$\top$
$\top$	$\bot$	$\top$	$\bot$	$\top$
$\bot$	$\top$	$\top$	$\bot$	$\top$
$\bot$	$\bot$	$\bot$	$\top$	$\bot$

Table 3. A truth table demonstrating that $p \vee q \equiv (p \mathbin{\overline{\vee}} q) \mathbin{\overline{\vee}} (p \mathbin{\overline{\vee}} q)$ for any propositions p and q. In order to see this, simply observe that the column labelled by $p \vee q$ is identical to the column labelled by $(p \mathbin{\overline{\vee}} q) \mathbin{\overline{\vee}} (p \mathbin{\overline{\vee}} q)$.

For conjunctions, we can take inspiration from the fact that $p \wedge q \equiv \neg(\neg p \vee \neg q)$, which was given to us in the problem statement. This is a particular application of *De Morgan's theorems*. Since we know how to translate $\neg$ and $\vee$ individually, we can just apply our translations to each individual $\neg$ and $\vee$ we see in the expression $\neg(\neg p \vee \neg q)$ to arrive at a translation equivalent to $p \wedge q$.

p	q	$p \wedge q$	$p \mathbin{\overline{\vee}} p$	$q \mathbin{\overline{\vee}} q$	$(p \mathbin{\overline{\vee}} p) \mathbin{\overline{\vee}} (q \mathbin{\overline{\vee}} q)$
$\top$	$\top$	$\top$	$\bot$	$\bot$	$\top$
$\top$	$\bot$	$\bot$	$\bot$	$\top$	$\bot$
$\bot$	$\top$	$\bot$	$\top$	$\bot$	$\bot$
$\bot$	$\bot$	$\bot$	$\top$	$\top$	$\bot$

Table 4. A truth table demonstrating that $p \wedge q$ is logically equivalent to the negation of $(p \mathbin{\overline{\vee}} p) \mathbin{\overline{\vee}} (q \mathbin{\overline{\vee}} q)$, which we compute by taking that expression and NOR-ing it with itself, for any propositions p and q. In order to see this, compare the first and last color-shaded columns and notice that they are identical. Along the way, notice that the intermediate column $p \mathbin{\overline{\vee}} p$ computes $\neg p$, and the column $q \mathbin{\overline{\vee}} q$ computes $\neg q$, so if we use each of these expressions as the input to $\overline{\vee}$, we will obtain $\neg((\neg p) \vee (\neg q))$.

For conditional statements, we again look to the problem statement for inspiration, recalling that $p \to q \equiv \neg p \vee q$. We take a similar approach to the previous connective: given two propositions p and q, we will first negate p, and then combine $\neg p$ with q using the construction for disjunctions we uncovered earlier.

p	q	$p \to q$	$p \mathbin{\overline{\vee}} p$	$((p \mathbin{\overline{\vee}} p) \mathbin{\overline{\vee}} q) \mathbin{\overline{\vee}} ((p \mathbin{\overline{\vee}} p) \mathbin{\overline{\vee}} q)$
$\top$	$\top$	$\top$	$\bot$	$\top$
$\top$	$\bot$	$\bot$	$\bot$	$\bot$
$\bot$	$\top$	$\top$	$\top$	$\top$
$\bot$	$\bot$	$\top$	$\top$	$\top$

Table 5. A truth table demonstrating that $p \to q$ is logically equivalent to $((p \mathbin{\overline{\vee}} p) \mathbin{\overline{\vee}} q) \mathbin{\overline{\vee}} ((p \mathbin{\overline{\vee}} p) \mathbin{\overline{\vee}} q)$ for any propositions p and q. In order to see this, notice that the first and last color-shaded columns are identical.

Finally, for biconditional statements, we would ideally rely on the important observation that $p \leftrightarrow q \equiv (p \to q) \wedge (q \to p)$ for any propositions p and q. We proceed as in the previous examples by combining our translations of $\wedge$ and $\to$ from before… or at least we'd like to; however, if we attempted that construction, we would arrive at the following expression as a translation for the biconditional.

$$\Big(\Big(\big((p \mathbin{\overline{\vee}} p) \mathbin{\overline{\vee}} q\big) \mathbin{\overline{\vee}} \big((p \mathbin{\overline{\vee}} p) \mathbin{\overline{\vee}} q\big)\Big) \mathbin{\overline{\vee}} \Big(\big((p \mathbin{\overline{\vee}} p) \mathbin{\overline{\vee}} q\big) \mathbin{\overline{\vee}} \big((p \mathbin{\overline{\vee}} p) \mathbin{\overline{\vee}} q\big)\Big)\Big) \mathbin{\overline{\vee}} \Big(\Big(\big((p \mathbin{\overline{\vee}} p) \mathbin{\overline{\vee}} q\big) \mathbin{\overline{\vee}} \big((p \mathbin{\overline{\vee}} p) \mathbin{\overline{\vee}} q\big)\Big) \mathbin{\overline{\vee}} \Big(\big((p \mathbin{\overline{\vee}} p) \mathbin{\overline{\vee}} q\big) \mathbin{\overline{\vee}} \big((p \mathbin{\overline{\vee}} p) \mathbin{\overline{\vee}} q\big)\Big)\Big)$$

While there's certainly nothing *technically* wrong with that expression—it *is* equivalent to $p \leftrightarrow q$—we could write a much shorter expression that says the same thing if we make one realization first. Recall that $p \to q \equiv \neg p \vee q$, and also recall that $p \wedge q \equiv (\neg p \vee \neg q)$. We can use the second fact—which is an instance of De Morgan's theorems—to turn the conjunction in $(p \to q) \wedge (q \to p)$ into a *negated disjunction* involving *negated conditionals*, and each of those negated conditionals is just a fancy *negated disjunction*. Every time we have a negated disjunction, that's just an instance of the $\overline{\vee}$ connective.

$$\begin{aligned}
(p \to q) \wedge (q \to p) &\equiv \neg\big(\neg(p \to q) \vee \neg(q \to p)\big) \\
&\equiv \neg\big(\neg(\neg p \vee q) \vee \neg(\neg q \vee p)\big) \\
&\equiv \neg\Big(\big(\neg p \mathbin{\overline{\vee}} q\big) \vee \big(\neg q \mathbin{\overline{\vee}} p\big)\Big) \\
&\equiv \big(\neg p \mathbin{\overline{\vee}} q\big) \mathbin{\overline{\vee}} \big(\neg q \mathbin{\overline{\vee}} p\big)
\end{aligned}$$

If we then apply our first translation to write $\neg p \equiv p \mathbin{\overline{\vee}} p$ and $\neg q \equiv q \mathbin{\overline{\vee}} q$, we obtain the much shorter expression $p \leftrightarrow q \equiv ((p \mathbin{\overline{\vee}} p) \mathbin{\overline{\vee}} q) \mathbin{\overline{\vee}} ((q \mathbin{\overline{\vee}} q) \mathbin{\overline{\vee}} p)$. We verify that this is correct with the truth table below.

p	q	$p \leftrightarrow q$	$(p \mathbin{\overline{\vee}} p) \mathbin{\overline{\vee}} q$	$(q \mathbin{\overline{\vee}} q) \mathbin{\overline{\vee}} p$	$((p \mathbin{\overline{\vee}} p) \mathbin{\overline{\vee}} q) \mathbin{\overline{\vee}} ((q \mathbin{\overline{\vee}} q) \mathbin{\overline{\vee}} p)$
$\top$	$\top$	$\top$	$\bot$	$\bot$	$\top$
$\top$	$\bot$	$\bot$	$\top$	$\bot$	$\bot$
$\bot$	$\top$	$\bot$	$\bot$	$\top$	$\bot$
$\bot$	$\bot$	$\top$	$\bot$	$\bot$	$\top$

Table 6. A truth table demonstrating that $p \to q$ is logically equivalent to $((p \mathbin{\overline{\vee}} p) \mathbin{\overline{\vee}} q) \mathbin{\overline{\vee}} ((p \mathbin{\overline{\vee}} p) \mathbin{\overline{\vee}} q)$ for any propositions p and q. In order to see this, notice that the first and last color-shaded columns are identical. The intermediate columns $(p \mathbin{\overline{\vee}} p) \mathbin{\overline{\vee}} q$ and $(q \mathbin{\overline{\vee}} q) \mathbin{\overline{\vee}} p$ correspond to $\neg(\neg p \to q)$ and $\neg(\neg q \to p)$ respectively.

With that, we have shown that any proposition involving the standard logical connectives can be rewritten so that it uses $\overline{\vee}$ as its sole logical connective.

3. Consider the infinite sequence of sentences indexed by natural numbers

$$\sigma_0, \sigma_1, \sigma_2, ..., \sigma_i, ...$$

where, for each natural number i, the ith sentence σ_i is defined as shown below.

$$\begin{aligned}\sigma_i &:= \text{“}\sigma_j \textit{ is false for every natural number } j \textit{ greater than } i.\text{”} \\ &\equiv \forall j\Big((j > i) \to \big(\sigma_j \equiv \bot\big)\Big)\end{aligned}$$

We will demonstrate that ***none*** of these sentences admits a consistent truth value. We will work under the temporary presumption that these sentences *do* have truth values and see where this leads us, beginning with the first sentence σ_0.

First, suppose σ_0 is *true.* That means that each of the sentences σ_1, σ_2, σ_3, … are individually *false.* In particular, this means that σ_1 is *false.* Since σ_1 said that all of the sentences after it are *false,* and we believe the *opposite* of what σ_1 says, we should now believe that *“not all of the sentences after* σ_1 *are* false.” That clearly implies that *there is* a sentence after σ_1 which is *true*—let’s call it σ_k, where $1 < k$ and $\sigma_k \equiv \top$. However, since $0 < 1 < k$, and since we believed what σ_0 said, we already knew $\sigma_k \equiv \bot$ because σ_0 told us that every sentence after it was *false.* Thus, σ_k is both *true* and *false,* which is clearly a contradiction. ↯
As a result, σ_0 cannot be *true.*

Now, towards the contrary, let’s suppose that σ_0 is *false.* Reading what σ_0 says, we should now believe that *“not every sentence after* σ_0 *is* false,” the negation of σ_0. This clearly means there is some sentence after σ_0 which is *true*—let’s call that sentence σ_ℓ, where $0 < \ell$ and $\sigma_\ell \equiv \top$. We can now essentially conduct the same analysis we initially did for σ_0. Since σ_ℓ is *true,* every sentence after σ_ℓ must be *false;* in particular, $\sigma_{\ell+1} \equiv \bot$. Since $\sigma_{\ell+1}$ is *false,* not every sentence after $\sigma_{\ell+1}$ is *false,* and so there must be some other sentence out there—after $\sigma_{\ell+1}$—which is *true.* Let’s call that sentence σ_m, where $\ell + 1 < m$ and $\sigma_m \equiv \top$. However, notice that $\ell < \ell + 1 < m$, and recall that we believe σ_ℓ when it says that every sentence after it is *false;* this implies $\sigma_m \equiv \bot$ in particular. So now, having deduced σ_m is both *true* and *false,* we have a contradiction. ↯
As a result, σ_0 cannot be *false.*

Therefore, we conclude that σ_0 cannot admit a single consistent truth value. Looking back over our argument, there is actually nothing special about σ_0 compared to any of the other sentences in the sequence. In fact, if we take the argument we just made and replace every instance of “0” and “1” with “i” and “$i + 1$,” where i represents an arbitrary natural number, we would have a completely coherent proof that σ_i has no truth value *for any natural number* i. Therefore, we conclude (for real this time) that ***none of the sentences in the infinite sequence admit a single consistent truth value.***